

Lecture 23.02: Types of Machine Learning: Supervised Learning

 Python 3.10+

 Scikit-Learn 1.4+

Status Completed

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1. Mathematical Formulation of Supervised Learning

In **Supervised Learning**, the learning algorithm is provided with a training dataset of N labeled examples:

$$\mathcal{D}_{\text{train}} = \left\{ (\mathbf{x}_i, y_i) \right\}_{i=1}^N$$

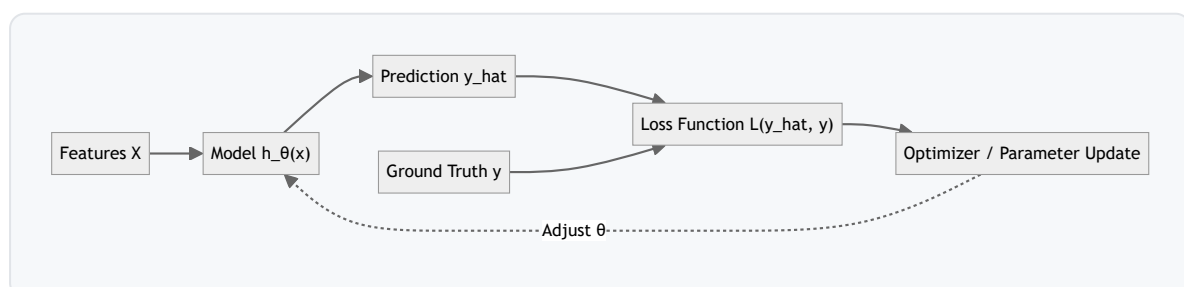
where:

- $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^d$ is a d -dimensional feature vector (independent variables).
- $y_i \in \mathcal{Y}$ is the true ground-truth target label (dependent variable).

Objective of the Learner:

The goal is to learn a hypothesis function $h_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ parameterized by θ such that it minimizes expected risk (loss) over unseen data:

$$\boldsymbol{\theta}^* = \mathop{\arg\min}_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(h_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i)$$



2. Core Components of Supervised Learning

Component	Notation	Description	Example (Insurance Dataset)
Instance / Sample	$\mathbf{x}_i$	A single observed entity	A specific insurance policyholder
Feature Vector	$[x_{i1}, \dots, x_{id}]^T$	Numerical/categorical attributes	[age, bmi, children, smoker]
Target Label	y_i	The true outcome being predicted	Annual healthcare charges
Model Output	$\hat{y}_i = h(\mathbf{x}_i)$	The model's prediction	Estimated charges
Supervisory Signal	$e_i = y_i - \hat{y}_i$	Discrepancy used for learning	Error used by Gradient Descent

3. Types of Supervised Learning Tasks

Supervised Learning is partitioned based on the nature of the target space $\mathcal{Y}$:

- Regression** ($\mathcal{Y} \subseteq \mathbb{R}$): Target is continuous.
 - Examples: Predicting insurance charges, house prices, temperature.
- Binary Classification** ($\mathcal{Y} \in \{0, 1\}$): Target is binary.
 - Examples: Patient has disease (1) or healthy (0), Email is Spam (1) or Ham (0).
- Multi-Class Classification** ($\mathcal{Y} \in \{1, \dots, K\}$): Target has discrete multiple classes.
 - Examples: Classifying vehicle type (Car, Truck, Motorcycle, Bus).

4. Python Implementation: Simulating the Supervised Learning Loop

```
import numpy as np

# Training features and targets
X = np.array([[1.0], [2.0], [3.0], [4.0]])
y = np.array([2.5, 4.5, 6.5, 8.5])

# Initialize parameter w (hypothesis:  $y_{\text{hat}} = w * x$ )
w = 0.0
lr = 0.05

print(f"{'Epoch':<6} | {'Weight w':<10} | {'MSE Loss':<12} | {'Predictions y_hat':<12}")
print("-" * 55)

for epoch in range(15):
    # Forward pass
    y_pred = X.dot(w).squeeze()
    # Compute loss (MSE)
    loss = np.mean((y_pred - y)**2)
    # Compute gradient:  $d\text{Loss}/dw = 2/N * \sum((y_{\text{pred}} - y) * x)$ 
    grad = np.mean(2 * (y_pred - y) * X.squeeze())
    # Update weight
    w = w - lr * grad

    if epoch % 3 == 0 or epoch == 14:
        print(f"{'epoch':<6} | {'w':<10.4f} | {'loss':<12.4f} | {'np.round(y_pred, 2)}")
```